

Q1: (a) Given: $f(x) = ax^2, 0 < x < 1$
 $= 0, \text{ elsewhere}$

Step 1: Finding a

Probability density function

$$f(x) = ax^2, 0 < x < 1$$

$$f(x) = 0, \text{ other cases}$$

Properties of PDF

① Non-Negative, $f(x) \geq 0$

② Total probability range to be equal to 1

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

So in our case $\Rightarrow \int_0^1 ax^2 dx = 1$

$$\Rightarrow a \int_0^1 x^2 dx = 1$$

Applying integration formula $\Rightarrow a \times \left(\frac{x^3}{3} \right)_0^1 = 1$

$$\Rightarrow a \times \frac{1}{3} = 1$$

Constant $a \Rightarrow \boxed{a = 3}$

Step 2°: finding distribution function $F(x)$

so: $F(x) = P(X \leq x)$

Let t be random variable

Case 1: $x \leq 0$, Since PDF rule is to lie b/w 0 to 1

$\therefore F(x) = 0, x \leq 0$

Case 2: $0 < x < 1$

$$F(x) = \int_0^x 3t^2 dt$$

$$\Rightarrow 3 \int_0^x t^2 dt = 3 \left(\frac{t^3}{3} \right)_0^x$$

$$\Rightarrow \left(t^3 \right)_0^x = \underbrace{x^3 - 0}_{\text{applying limits}} = x^3$$

hence, $F(x) = x^3, 0 < x < 1$

Case 3: $x \geq 1$; Since total probability is 1

$$F(x) = 1, x \geq 1$$

So final distribution function:

Answer: $F(x) = \begin{cases} 0, & x \leq 0 \\ x^3, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$

Calculating Mean: Mean for

Continuous
random
Variables

PRATEEK SRIVASTAVA

BIT SID: 2025AC05601

Section 4

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$$E(x) = \int x f(x) dx$$

Now substituting the PDF

$$E(x) = \int_0^1 x (3x^2) dx$$

$$= 3 \int_0^1 x^3 dx = 3 \left(\frac{x^4}{4} \right)_0^1$$

$$= 3 \left(\frac{1}{4} - 0 \right) = \frac{3}{4}$$

$$\text{So: } \boxed{E(x) = \frac{3}{4}} \leftarrow \text{Mean}$$

Calculating Variance: $E(x^2) = \int x^2 f(x) dx$

So, if we substitute PDF

$$E(x^2) = \int_0^1 x^2 (3x^2) dx$$

$$= 3 \int_0^1 x^4 dx = 3 \times \left(\frac{x^5}{5} \right)_0^1$$

$$= 3 \left(\frac{1}{5} - 0 \right) = \frac{3}{5}$$

Using variance formula:

$$\text{Var}(x) = E(x^2) - [E(x)]^2$$

$$\text{Var}(x) = \frac{3}{5} - \left(\frac{3}{4}\right)^2 = \frac{3}{5} - \frac{9}{16} = \frac{48}{80} - \frac{45}{80}$$

$$\text{Var}(x) = \frac{3}{80}$$

Question 2:

Probability of survival of patient = 90%

so, $p = .9$

Probability of death (failure) = $1 - .9 = \underline{\underline{.1}}$

so, $q = .1$, $n = 100$

Let x (no. of patients who survive

- Also since number of trials is fixed and
- outcomes for each patient is fixed to 2 (survive or death)
- Probability of success is constant p
- events are independent

all this above satisfies conditions of

Binomial Distribution

Therefore $X \sim B(n, p)$

$$X \sim B(100, .9)$$

Since calculating separate probabilities is lengthy
→ let's use Normal approximation to Binomial distribution

$$\begin{array}{l|l} \frac{np \geq 5}{100 \times .9 = 90} & \frac{nq \geq 5}{100 \times .1 = 10} \\ 90 \geq 5 & 10 \geq 5 \end{array}$$

So, Normal approximation is applicable.

Calculating Mean: $\mu = np = 100 \times .9 = 90$

Calculating Standard deviation: $\sigma = \sqrt{npq}$

$$\text{Variance: } \sigma^2 = 100 \times .9 \times .1 = 9$$

$$\sigma = \sqrt{9} = 3$$

Standard deviation

$$\sigma = 3$$

Now: Applying Continuity Correction X ranges from

$$83.5 < X < 95.5$$

So Probability: $P(83.5 < X < 95.5)$

Now Converting to Z scores: $Z = \frac{X - \mu}{\sigma}$ Section 4

$$Z_1 = \frac{83.5 - 90}{3} = -\frac{6.5}{3} = -2.17$$

$$Z_2 = \frac{95.5 - 90}{3} = \frac{5.5}{3} = 1.83$$

Now: $P(-2.17 < Z < 1.83)$

Using standard Normal table

$$P(Z < 1.83) = .9664$$

$$\text{and } P(Z < -2.17) = .0150$$

So. Required probability $\Rightarrow P(Z < 1.83) - P(Z < -2.17)$

$$= .9664 - 0.0150$$

$$= .9514$$

$$\text{So, } \boxed{P(84 \leq X \leq 95) = .9514}$$

or 95.14%

Answer: hence 95.14% Chance that the number of patients surviving the operation lie b/w 84 & 95.

Question 3:

Box 1 10 bulbs 4 defective

Box 2 6 bulbs 1 defective

Box 3 8 bulbs 3 defective

Answer:

Part (a): Probability of selecting boxes.

(a) Since no box is favoured then probability of selecting each box is equal so:

$$P(B_1) = P(B_2) = P(B_3) = 1/3 \leftarrow$$

(b) Selected bulb defective for box 1

$$P(D | \text{Box 1}) = \frac{4}{10} = \frac{2}{5} = .4 \leftarrow$$

(c) Selected bulb defective for box 2

$$P(D | \text{Box 2}) = 1/6 = .167 \leftarrow$$

(d) Selected bulb defective for box 3

$$P(D | \text{Box 3}) = \frac{3}{8} = .375 \leftarrow$$

(c) If the selected bulb is defective.

3 possible ways of selecting defective bulb

1. Box 1 2. Box 2 3. Box 3

Let's find total probability of picking defective.

$$P(D) = P(B_1) \cdot P(D|B_1) + P(B_2) \cdot P(D|B_2) + P(B_3) \cdot P(D|B_3)$$

$$P(D) = \frac{1}{3} \times \frac{2}{5} + \frac{1}{3} \times \frac{1}{6} + \frac{1}{3} \times \frac{3}{8}$$

$$= \frac{2}{15} + \frac{1}{18} + \frac{1}{8} = \frac{48 + 20 + 45}{360} \leftarrow \text{LCM}$$

$$P(D) = \frac{113}{360} = \underline{.314} \quad 31.4\% \text{ Chance of selecting defective bulb}$$

Now using formula of Bayes's theorem

$$P(B_n|D) = \frac{P(B_n) P(D|B_n)}{P(D)}$$

1) Bulb defective from Box 1: $P(B_1|D) = \frac{P(B_1) P(D|B_1)}{P(D)} = \frac{\frac{1}{3} \times \frac{2}{5}}{\frac{113}{360}} = \frac{48}{113} = \underline{.425}$
hence 42.5% Chance.

2) Bulb defective from Box 2: $P(B_2|D) = \frac{P(B_2) P(D|B_2)}{P(D)} = \frac{\frac{1}{3} \times \frac{1}{6}}{\frac{113}{360}} = \frac{20}{113} = \underline{.177}$
hence 17.7% Chance

3) Bulb defective from Box 3: $P(B_3|D) = \frac{P(B_3) P(D|B_3)}{P(D)} = \frac{\frac{1}{3} \times \frac{3}{8}}{\frac{113}{360}} = \frac{45}{113} = \underline{.398}$
hence 39.8% Chance

Just to validate: $\frac{48 + 20 + 45}{113} = 1$